

CIVICPULSE AI

Turning citizen voices into intelligent city action

A streetlight, reported three times

Asha reported the dark streetlight near her bus stop three times in two weeks. Each report sat unranked in a 60-ticket queue.

(Illustrative scenario)

126,974
BBMP grievances in ~5.5 months

31.4%
Streetlights of total complaints

58%
Road maintenance resolved

Source: OpenCity analysis of BBMP data, Jul 2025.

Personas: Citizen (Asha) · Ward engineer (Kavitha)

Independent civic-tech prototype · Not affiliated with government

Reports pile up. Nothing gets ranked.

CivicPulse turns every report into ranked, located action.

TODAY: FIRST-COME QUEUES

- **Duplicate inflation**

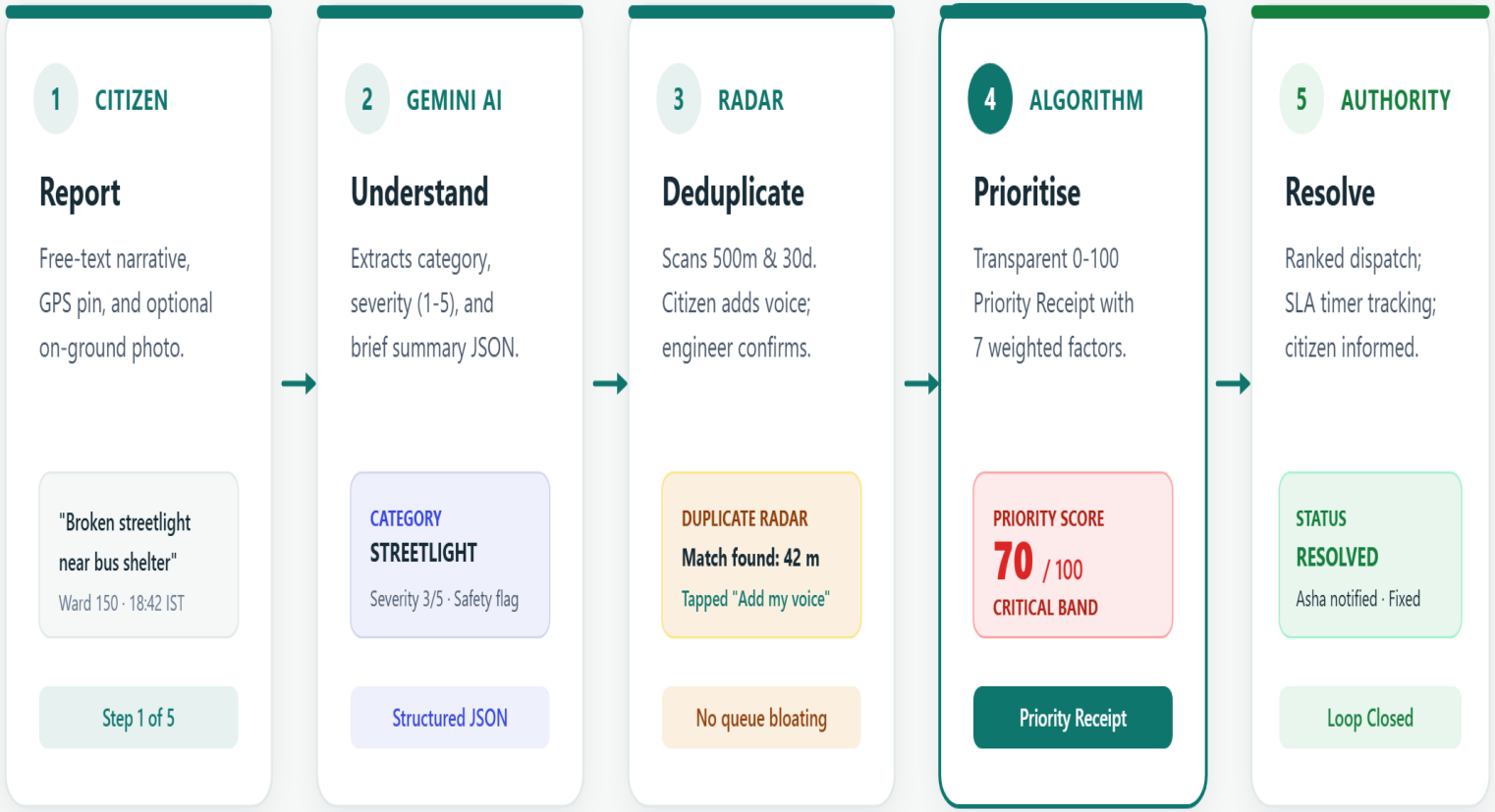
  Same broken light filed repeatedly, bloating unranked backlogs.

- **Blind queues**

  Critical public hazards wait buried behind cosmetic tasks.

- **Citizen silence**

  No updates sent back, destroying trust and triggering re-reports.



AI suggests. Code decides. People confirm.

Transparent civic triage with zero black-box scoring.

1. Gemini AI Reads
Advisory understanding

Extracts category, summary & hazards. Strict JSON output. Keyword fallback guarantees 100% uptime if offline.

2. Duplicate Radar
Nearby similarity

Scans 500 m & 30 days via embeddings. Citizens tap "Add my voice"; engineers confirm links. Zero auto-merging.

3. Priority Receipt
Auditable deterministic code

Itemised 0-100 score from 7 factors. Formulaic TypeScript (§29). Gemini has zero weight in the score.

CIVICPULSE PRIORITY RECEIPT §29
REF: CP-BLR-2026-0842 · WARD 150 (BELLANDUR) · WORKED EXAMPLE E1

CRITICAL

CATEGORY
DRAINAGE (OPEN DRAIN NEAR SCHOOL)

TOTAL PRIORITY SCORE
70 / 100

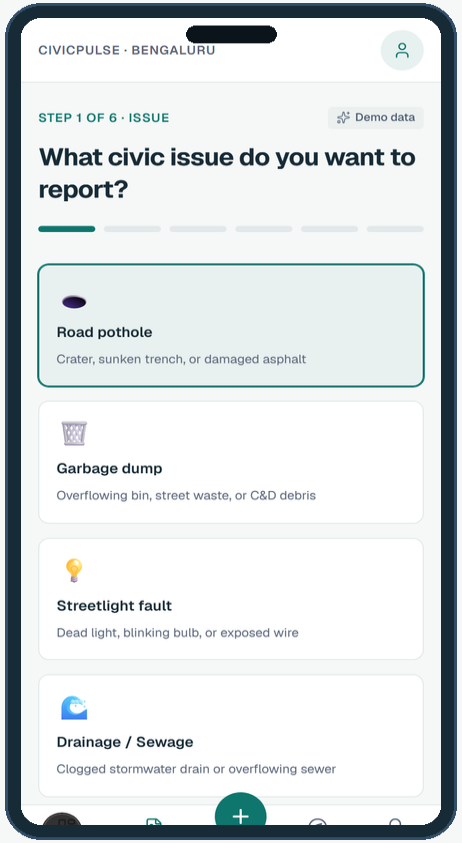
FACTOR KEY & WEIGHT	GROUND-TRUTH DETAIL & CONDITIONS	CALCULATED CONTRIBUTION
Severity (S) · 25%	Base severity 4 of 5 (factor 0.75)	**+18.75 pts**
Safety Hazard (H) · 20%	Risk reported: INJURY_RISK (factor 1.0)	**+20.00 pts**
Reports (R) · 15%	4 independent reporters (u=3, ln(4)/ln(11) = 0.578)	**+8.67 pts**
Waiting Time (D) · 15%	20 h elapsed of 48 h SLA target (factor 0.208)	**+3.13 pts**
Affected Scale (A) · 10%	School / arterial pedestrian zone (scale 1.0)	**+10.00 pts**
Hotspot Density (G) · 10%	4 other open complaints within 500 m (factor 0.40)	**+4.00 pts**
Season (T) · 5%	Monsoon season sensitive (November, factor 1.0)	**+5.00 pts**

Pre-rounding sum: 69.55 pts → roundHalfUp() = 70
Deterministic TypeScript (§29 pure function) · Gemini has zero scoring weight

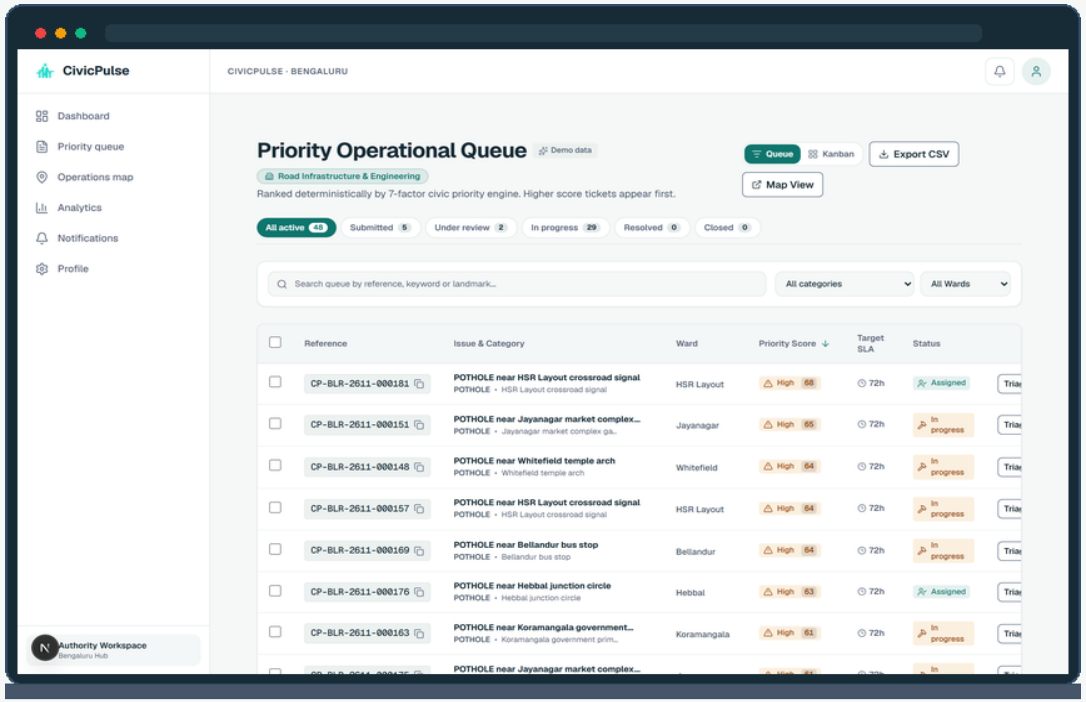
FINAL: 70 [CRITICAL]

It works today

A verified prototype deployed on production free tiers.



Mobile Citizen Wizard
AI suggestion & location



Authority Triage Queue · Ranked 0-100 tickets with SLA countdowns

Citizen reports → AI analyses → Duplicate linked → Engineer resolves → Citizen notified

civicpulse.vercel.app

Judge Access: 1-Click Demo Roles

Citizen · Roads Authority · Admin

Pre-seeded with 240 active reports

Built free, built to scale, built honestly

Open civic technology on ₹0 infrastructure.

FRONTEND
Next.js 16
React 19 · Tailwind v4
Geist Sans & Radix UI

BACKEND & DATA
Firebase Spark
Auth + Firestore Admin
Free Tier · 0 client writes

AI REASONING
Gemini 2.5 Flash
Google AI Studio Free
Deterministic fallback

GEOSPATIAL TILES
OpenFreeMap
MapLibre GL + DataMeet
100% Free · No API key

DEPLOYMENT
Vercel Hobby
Edge + Serverless
₹0 / month total

OPEN DATA SOURCES & LICENCES

- **BBMP Grievance Records:** OpenCity.in analysis (Jul 2025)
- **Bengaluru Ward Boundaries:** DataMeet Community Maps (CC BY 4.0)
- **Base Vector Maps:** OpenFreeMap / OpenStreetMap (ODbL)
- **Synthetic Evaluation Data:** 240 seeded reports labelled 'Demo data'

PROTOTYPE IMPACT & NEXT STEPS

240
Reports analysed

18%
Duplicates averted

< 3s
Median triage time

Labelled Prototype data · Single-ward municipal pilot with Hindi & Kannada support.



Reported once. Prioritised. Fixed. And Asha knew.

CivicPulse AI · INNOVATE Online Hackathon 2026 by Aveeranex on Unstop